

Software Systems - Assignment 2

Part A — Model Creation Task

System Description: Smart Campus Navigation App

A university plans to develop a **Smart Campus Navigation App** that helps students and visitors find buildings, classrooms, offices, and other campus facilities. The app should display routes, estimated walking times, and accessibility options (like ramps or elevators). It should also allow users to search for a faculty member or event location and get directions to them.

Tasks:

1. **Use Case Diagram:**

Draw a *Use Case Diagram* that captures the main interactions between the user (student/visitor) and the system.

2. **Detailed Use Cases:**

Write *detailed descriptions* for at least **three key use cases** — for example:

- a. “Search for a Building”
- b. “Get Directions”
- c. “Find Faculty Office”

Each use case description should include:

- a. Actors involved
- b. Preconditions
- c. Main flow of events
- d. Alternative/exception flows
- e. Postconditions

3. **Requirements Model (Class Diagram):**

Draw a *Class Diagram* representing the main classes, their attributes, and relationships.

For example, include classes such as User, Building, Map, Route, and Location, and show associations like “User requests Route” or “Route includes Locations”.

Part B — Model Correction Task

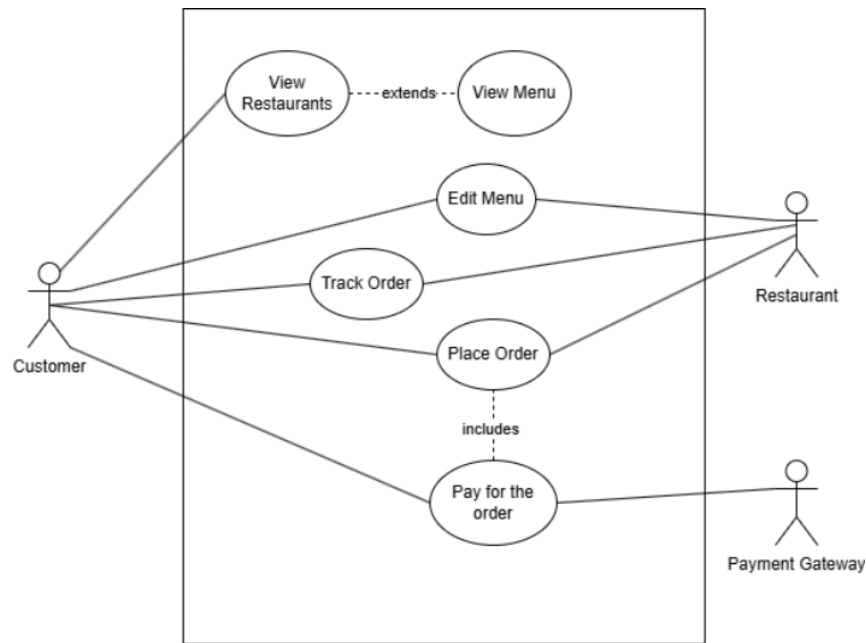
System Description: Online Food Ordering System

An online platform allows users to browse restaurants, view menus, place orders, and make online payments. Restaurants can update their menus, confirm or reject orders, and view order history. However, the following requirements model (below) contains **mistakes, missing elements, and ambiguities**.

Tasks:

1. Identify **all the mistakes, missing cases and ambiguities** in the given models.
2. Redraw the **Use Case Diagram**, **rewrite the Use Case descriptions**, and **recreate the Class Diagram** correctly.
3. Clearly explain **what corrections** you made and **why** (in 3–5 bullet points per model).

Use Case Diagram



Detailed Use Cases

Use Case 1: View Restaurants

Element	Description
Actors	Customer
Preconditions	Customer is logged in.
Main Flow	1. Customer opens “Browse Restaurants” page. 2. System displays a list of restaurants. 3. Customer applies filters (cuisine, price, rating). 4. System sorts results alphabetically.
Alternate Flow	(a) If no restaurants match, system displays “No Results”.
Postconditions	The system saves the filtered list for future sessions.

Use Case 2: View Menu

Element	Description
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Actors	Customer, Restaurant
Preconditions	Customer has selected a restaurant.
Main Flow	1. System requests the menu from Restaurant. 2. Restaurant uploads the latest menu file.3. System displays menu items.
Alternate Flow	(a) If restaurant is offline, system shows “Service Unavailable.”
Postconditions	Menu is cached on the customer’s device for future use.

Use Case 3: Make Payment

Element	Description
Actors	Customer, Payment Gateway
Preconditions	Order has been placed but not confirmed by restaurant.
Main Flow	1. Customer selects payment method.2. System redirects to Payment Gateway.3. Gateway authorizes and deducts payment.4. System stores payment token.
Alternate Flow	(a) If authorization fails, system cancels the order.
Postconditions	Order status marked “Paid.”

Use Case 4: Confirm Order (Restaurant-side)

Element	Description
Actors	Restaurant
Preconditions	Restaurant system is online.
Main Flow	1. Restaurant views new orders.2. Restaurant clicks “Confirm.”3. System notifies the Customer.
Postconditions	Order status updated accordingly.

Requirements Model (Class Diagram)

Online Food Ordering System - Flawed Requirements Model

